# 02 · Etiquetado por reglas con PySpark: intención, categoría, urgencia y frustración

**Objetivo:** leer el archivo Parquet generado por el notebook 01 directamente con Spark ( `spark.read.parquet` ), asignar a cada tweet de **cliente** sus etiquetas automáticas mediante funciones nativas de PySpark ( `F.when` , `F.rlike` , `F.greatest` ) y exportar tanto el dataset completo etiquetado en formato Parquet como la muestra aleatoria estratificada para validación manual.

| Archivo generado | Para qué sirve |
|---|---|
| `../data/processed/clientes_etiquetados.parquet` | **Todos** los tweets de clientes con la etiqueta automática de las reglas. |
| `../data/processed/muestra_para_etiquetar.csv` | Muestra estratificada **sin** etiquetas automáticas, para etiquetar a mano a ciegas. |
| `../data/processed/muestra_clave_reglas.csv` | Los mismos tweets de la muestra **con** la etiqueta automática; se usa al final para calcular el F1-score. |

## 1. Preparación del entorno y lectura directa de Parquet con Spark

Inicializamos la sesión de Spark 3.5.9 siguiendo la configuración de los laboratorios y leemos `apple_support_filtrado.parquet` directamente como DataFrame de Spark.

In [2]:
```
# ============================================================
# 1. PREPARACIÓN DE SPARK Y LECTURA DE PARQUET
# ============================================================
import os
import sys
from pathlib import Path
import pandas as pd
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics import classification_report

from pyspark.sql import SparkSession
from pyspark.sql import functions as F
from pyspark import StorageLevel

python_executable = sys.executable
os.environ["PYSPARK_PYTHON"] = python_executable
os.environ["PYSPARK_DRIVER_PYTHON"] = python_executable
os.environ["SPARK_LOCAL_IP"] = "127.0.0.1"

RUTA_IN = Path("../data/processed/apple_support_filtrado.parquet")
RUTA_OUT = Path("../data/processed/clientes_etiquetados.parquet")
RUTA_MUESTRA = Path("../data/processed/muestra_para_etiquetar.csv")
RUTA_CLAVE = Path("../data/processed/muestra_clave_reglas.csv")

assert RUTA_IN.exists(), f"No se encontró el archivo Parquet de entrada: {RUTA_IN.r


spark = (
    SparkSession.builder
    .appName("Tesis_02_Clasificacion_Reglas")
    .master("local[*]")
    .config("spark.driver.host", "127.0.0.1")
    .config("spark.driver.bindAddress", "127.0.0.1")
    .config("spark.sql.shuffle.partitions", "8")
    .config("spark.sql.parquet.int96RebaseModeInRead", "CORRECTED")
    .config("spark.sql.parquet.datetimeRebaseModeInRead", "CORRECTED")
```

```
    .config("spark.sql.legacy.parquet.nanosAsLong", "true")
    .getOrCreate()
)

spark.sparkContext.setLogLevel("WARN")
print("Spark:", spark.version, "| Spark UI:", spark.sparkContext.uiWebUrl)

# Lectura directa del Parquet en Spark (estilo Laboratorio 7 y 8)
df_filtrado = spark.read.parquet(str(RUTA_IN))
df_filtrado.printSchema()

df_clientes = (
    df_filtrado
    .filter(F.col("rol") == "cliente")
    .select("tweet_id", "rol", "fecha", "texto_limpio")
)
print("Total de tweets de clientes:", f"{df_clientes.count():,}")
```

```
Spark: 3.5.9 | Spark UI: http://127.0.0.1:4041
root
 |-- tweet_id: string (nullable = true)
 |-- rol: string (nullable = true)
 |-- author_id: string (nullable = true)
 |-- fecha: timestamp_ntz (nullable = true)
 |-- text: string (nullable = true)
 |-- texto_limpio: string (nullable = true)
 |-- n_palabras: integer (nullable = true)
 |-- in_response_to_tweet_id: string (nullable = true)

Total de tweets de clientes: 126,113
```

## 2. Catálogo de expresiones regulares

Cada lista de patrones se compila en una expresión regular case-insensitive ( `(?i)\b(?:...)` ) compatible con el motor `rlike` de Spark SQL.

In [3]:

```
# ============================================================
# 2. CATÁLOGO DE EXPRESIONES REGULARES PARA SPARK (RLIKE)
# ============================================================
FALLA_FUERTE = [
    r"not working", r"stopped working", r"not able", r"unable", r"won't?t", r"can't?t
    r"cannot", r"couldn't?t", r"no longer", r"broken", r"crash", r"bug", r"error",
    r"stuck", r"freez", r"frozen", r"glitch", r"lag", r"fail", r"outage", r"down\b"
    r"slow", r"drain", r"dying", r"died", r"dies\b", r"lost", r"missing", r"disappe
    r"black screen", r"unresponsive", r"soft reset", r"hard reset", r"force restart
    r"fallen", r"fell\b", r"cracked", r"overheat", r"ruin",
    r"(?:isn|aren|wasn|weren|doesn|don|didn|hasn|haven|hadn)'?t\s+(?:\w+\s+){0,2}?"
    r"(?:allow\w*|let\w*|work\w*|respond\w*|load\w*|show\w*|charg\w*|connect\w*|syn
    r"not\s+(?:receiving|sending|syncing|charging|loading|showing|allowing|letting|
]

FALLA_DEBIL = [
    r"issue", r"problem", r"fix", r"restart", r"reboot", r"reset", r"keeps",
    r"annoying", r"weird", r"wrong"
]

SOLICITUD_EXPLICITA = [
    r"how (?:do|can|to|does|should)", r"is it possible", r"is there a way"
]

SOLICITUD = [
    r"where (?:do|can|is|are)", r"what (?:is|does|are|happens)", r"when (?:will|is|
    r"is there", r"does (?:anyone|apple|it|the)", r"do (?:you|apple)", r"can (?:i|y
    r"could (?:you|i)", r"any (?:way|idea|tips?|update|news)", r"please (?:help|adv
    r"how much", r"which", r"wondering", r"want to know", r"looking for", r"need (?
    r"i want to", r"would like to", r"trying to"
]

CATEGORIAS = {
    "telefonia_llamadas": [
        r"(?:phone |voice )?calls?\b", r"phone calls?", r"voicemail",
```

```
        r"dial", r"ringing", r"no service", r"\bcaller"
    ],
    "cuenta_icloud": [
        r"icloud", r"apple id", r"password", r"log ?in", r"sign ?in", r"locked",
        r"verification", r"2fa", r"backup", r"storage", r"account"
    ],
    "pagos_facturacion": [
        r"charged", r"charges?\b", r"billing", r"\bbills?\b", r"refund",
        r"payment", r"subscription", r"invoice", r"apple pay",
        r"credit card", r"purchase"
    ],
    "servicios_apple": [
        r"apple music", r"imessage", r"facetime", r"siri", r"icloud mail",
        r"apple tv", r"podcast", r"maps\b", r"itunes", r"airdrop",
        r"homepod", r"carplay"
    ],
    "conectividad": [
        r"wi-?fi", r"bluetooth", r"cellular", r"signal", r"network",
        r"mobile data", r"data\b", r"sim\b", r"carrier", r"roaming",
        r"hotspot", r"lte\b", r"internet"
    ],
    "apps_funciones": [
        r"app store", r"apps?\b", r"download", r"testflight", r"safari",
        r"mail\b", r"photos?\b", r"widget", r"notification", r"feature",
        r"quick edit", r"edit\b"
    ],
    "ios_software_rendimiento": [
        r"ios(?:\s?\d+)?\b", r"update", r"upgrade", r"macos", r"beta",
        r"install", r"firmware", r"keyboard", r"glitch", r"bug",
        r"freez", r"frozen", r"crash", r"lag", r"slow",
        r"unresponsive", r"performance"
    ],
    "hardware_dispositivo": [
        r"battery", r"screens?\b", r"charging", r"charger", r"cracked",
        r"fallen", r"fell\b", r"dropped", r"water", r"speaker",
        r"microphone", r"buttons?\b", r"camera", r"overheat",
        r"turn(?:ing)? on", r"display", r"touch id", r"face id"
    ]
}

DISPOSITIVO = [
    r"iphone", r"ipad", r"ipod", r"mac(?:book)?\b", r"imac", r"phone",
    r"(?:apple )?watch\b", r"airpods", r"device", r"laptop", r"tablet"
]

CRITICA = [
    r"outage", r"down for", r"(?:is|are) down\b", r"services? (?:is |are )?down",
    r"completely (?:dead|broken|unusable|died)", r"unusable", r"bricked",
    r"won'?t (?:turn on|power|boot|start)", r"boot ?loop", r"stuck on (?:the )?(?:a
    r"data loss", r"lost (?:all|everything|my (?:photos|data|contacts|messages))",
    r"(?:photos|data|contacts|messages) (?:gone|deleted|disappeared|lost)",
    r"hacked", r"unauthori[sz]ed", r"stolen", r"fraud", r"locked out",
    r"(?:can'?t|cannot|unable to|not able to)\s+(?:log ?in|sign ?in|access)",
    r"charged (?:twice|\d+)", r"double charged", r"disabled", r"no service", r"911"
    r"(?:not able to|unable to|can'?t|cannot|couldn'?t)\s+(?:\w+\s+){0,3}?(?:make|r
    r"(?:fell|fallen|dropped|water|wet)[^.!?]{0,40}(?:stopped working|dead|not work
]

MEDIA_EXTRA = [
    r"problem", r"issue", r"fix", r"keeps", r"restart", r"reboot", r"reset"
]

AMPLIFICADORES = [
    r"urgent", r"asap", r"immediately", r"still\b", r"for (?:days|hours|weeks|month
    r"since (?:yesterday|monday|tuesday|wednesday|thursday|friday|this morning|last
    r"(?:past|last) (?:\w+ ){0,2}(?:hours?|days?|weeks?)", r"twice", r"(?:three|fou
    r"again", r"constantly", r"every (?:time|day|hour)", r"all the time", r"everyon
    r"anyone else", r"all (?:users|of us)", r"for work", r"work(?:ing)? from", r"bu
]

FRUSTRACION = [
    r"fuck", r"shit", r"damn", r"crap", r"ridiculous", r"unacceptable",
    r"worst", r"horrible", r"frustrat", r"angry", r"switch(?:ing)? to",
```

```
    r"never (?:buy|again)", r"😡", r"😡", r"😤"
]

def patron(lista):
    return r"(?i)\b(?:" + "|".join(lista) + r")"

RX_FALLA_FUERTE = patron(FALLA_FUERTE + CRITICA)
RX_FALLA_DEBIL = patron(FALLA_DEBIL)
RX_SOL_EXPLICITA = patron(SOLICITUD_EXPLICITA)
RX_SOLICITUD = patron(SOLICITUD)
RX_CRITICA = patron(CRITICA)
RX_MEDIA = patron(FALLA_FUERTE + MEDIA_EXTRA)
RX_AMPL = patron(AMPLIFICADORES)
RX_FRUSTRACION = patron(FRUSTRACION)
RX_DISPOSITIVO = patron(DISPOSITIVO)
```

# 3. Función de clasificación distribuida en PySpark

Construimos las columnas `intencion`, `categoria`, `urgencia` y `frustracion` utilizando únicamente transformaciones nativas de Spark (`F.when`, `F.rlike`, `F.greatest`), evitando UDFs en Python para que Catalyst optimice el plan físico.

In [4]:
```
# ============================================================
# 3. CLASIFICACIÓN CON FUNCIONES NATIVAS DE SPARK (F)
# ============================================================
def clasificar_spark(df, col_texto="texto_limpio"):
    t = F.regexp_replace(F.coalesce(F.col(col_texto), F.lit("")), r"[’‘`]", "'")
    df_temp = df.withColumn("_t", t)

    # 1. Intención
    cond_fuerte = F.col("_t").rlike(RX_FALLA_FUERTE)
    cond_explicita = F.col("_t").rlike(RX_SOL_EXPLICITA)
    cond_debil = F.col("_t").rlike(RX_FALLA_DEBIL)
    cond_solicitud = F.col("_t").rlike(RX_SOLICITUD) | F.col("_t").contains("?")

    col_intencion = (
        F.when(cond_fuerte, F.lit("falla"))
        .when(cond_explicita, F.lit("solicitud"))
        .when(cond_debil, F.lit("falla"))
        .when(cond_solicitud, F.lit("solicitud"))
        .otherwise(F.lit("otro"))
    )
    df_temp = df_temp.withColumn("intencion", col_intencion)

    # 2. Categoría (conteo de coincidencias por categoría; desempate por orden del
    cols_puntaje = []
    for cat, lista_patrones in CATEGORIAS.items():
        expr_suma = F.lit(0)
        for p in lista_patrones:
            expr_suma = expr_suma + F.when(F.col("_t").rlike(patron([p])), 1).other
        nombre_col = f"_score_{cat}"
        df_temp = df_temp.withColumn(nombre_col, expr_suma)
        cols_puntaje.append((cat, nombre_col))

    df_temp = df_temp.withColumn(
        "_max_score",
        F.greatest(*[F.col(c) for _, c in cols_puntaje])
    )

    expr_cat = F.when(
        F.col("_t").rlike(RX_DISPOSITIVO),
        F.lit("dispositivo_general")
    ).otherwise(F.lit("otros"))

    for cat, nombre_col in reversed(cols_puntaje):
        expr_cat = F.when(
            (F.col("_max_score") > 0) & (F.col(nombre_col) == F.col("_max_score")),
            F.lit(cat)
        ).otherwise(expr_cat)
```

```
    df_temp = df_temp.withColumn("categoria", expr_cat)

    # 3. Urgencia (solo aplica para intención == 'falla')
    es_falla = F.col("intencion") == "falla"
    cond_critica = F.col("_t").rlike(RX_CRITICA)
    cond_media = F.col("_t").rlike(RX_MEDIA)
    cond_ampl = F.col("_t").rlike(RX_AMPL)

    col_urgencia = (
        F.when(~es_falla, F.lit("baja"))
        .when(cond_critica | (cond_media & cond_ampl), F.lit("critica"))
        .when(cond_media, F.lit("media"))
        .otherwise(F.lit("baja"))
    )
    df_temp = df_temp.withColumn("urgencia", col_urgencia)

    # 4. Frustración
    df_temp = df_temp.withColumn("frustracion", F.col("_t").rlike(RX_FRUSTRACION))

    cols_borrar = ["_t", "_max_score"] + [c for _, c in cols_puntaje]
    return df_temp.drop(*cols_borrar)
```

# 4. Prueba con ejemplos de verificación

Verificamos el comportamiento de las reglas sobre casos reales y de contraste antes de procesar todo el conjunto de datos.

In [5]:

```
# ============================================================
# 4. PRUEBA CON EJEMPLOS
# ============================================================
ejemplos = [
    ("@AppleSupport my phone is not able to receive or make any phone calls.",),
    ("@AppleSupport iphone glitch",),
    ("@AppleSupport mac glitch",),
    ("I've had to soft reset my phone twice in the past two hours because it became
    ("@AppleSupport please fix the fucking problem with keyboard or I'll switch to
    ("@AppleSupport since yesterdays update, my phone is dying of glitches and cons
    ("@AppleSupport It‘s fallen down and then stopped working",),
    ("Hey @AppleSupport the iOS update is ruining my Apple iPhone experience pls fi
    ("@AppleSupport No. And it also isn't allowing me to do the new quick edit thin
    ("@AppleSupport how do I restart my iPhone?",),
    ("@AppleSupport thank you so much, that worked!",)
]

df_ejemplos = spark.createDataFrame(ejemplos, ["texto_limpio"])
display(clasificar_spark(df_ejemplos).toPandas())
```

| | texto_limpio | intencion | categoria | urgencia | frustracion |
|---|---|---|---|---|---|
| **0** | @AppleSupport my phone is not able to receive ... | falla | telefonia_llamadas | critica | False |
| **1** | @AppleSupport iphone glitch | falla | ios_software_rendimiento | media | False |
| **2** | @AppleSupport mac glitch | falla | ios_software_rendimiento | media | False |
| **3** | I've had to soft reset my phone twice in the p... | falla | ios_software_rendimiento | critica | False |
| **4** | @AppleSupport please fix the fucking problem w... | falla | ios_software_rendimiento | media | True |
| **5** | @AppleSupport since yesterdays update, my phon... | falla | ios_software_rendimiento | critica | False |
| **6** | @AppleSupport It‘s fallen down and then stoppe... | falla | hardware_dispositivo | critica | False |
| **7** | Hey @AppleSupport the iOS update is ruining my... | falla | ios_software_rendimiento | media | False |
| **8** | @AppleSupport No. And it also isn't allowing m... | falla | apps_funciones | media | False |
| **9** | @AppleSupport how do I restart my iPhone? | solicitud | dispositivo_general | baja | False |
| **10** | @AppleSupport thank you so much, that worked! | otro | otros | baja | False |

# 5. Aplicación al dataset completo, plan físico y distribuciones

Aplicamos las reglas al DataFrame de clientes, inspeccionamos el plan físico con `explain("formatted")`, persistimos el resultado y calculamos las distribuciones con agregaciones de Spark.

In [6]:
```
# ============================================================
# 5. APLICACIÓN AL DATASET COMPLETO Y DISTRIBUCIONES EN SPARK
# ============================================================
df_etiquetado = clasificar_spark(df_clientes)
df_etiquetado.persist(StorageLevel.MEMORY_AND_DISK)

print("PLAN FÍSICO DE CLASIFICACIÓN:")
df_etiquetado.explain("formatted")

total_clientes = df_etiquetado.count()
print("Total de tweets de clientes etiquetados:", f"{total_clientes:,}")

for dim in ["intencion", "categoria", "urgencia", "frustracion"]:
    print(f"\n=== DISTRIBUCIÓN: {dim.upper()} ===")
    (
        df_etiquetado
        .groupBy(dim)
        .agg(F.count("*").alias("count"))
        .withColumn("pct", F.round(F.col("count") / F.lit(total_clientes), 3))
        .orderBy(F.desc("count"))
        .show(truncate=False)
    )

print("\n=== TABLA CRUZADA: CATEGORÍA vs URGENCIA (SOLO FALLAS) ===")
(
    df_etiquetado
    .filter(F.col("intencion") == "falla")
    .groupBy("categoria")
    .pivot("urgencia", ["baja", "media", "critica"])
    .count()
    .na.fill(0)
```

```
    .orderBy("categoria")
    .show(truncate=False)
)
```

```
PLAN FÍSICO DE CLASIFICACIÓN:
== Physical Plan ==
AdaptiveSparkPlan (10)
+- InMemoryTableScan (1)
      +- InMemoryRelation (2)
            +- * Project (9)
               +- * Project (8)
                  +- * Project (7)
                     +- * Project (6)
                        +- * Filter (5)
                           +- * ColumnarToRow (4)
                              +- Scan parquet  (3)


(1) InMemoryTableScan
Output [8]: [tweet_id#0, rol#1, fecha#3, texto_limpio#5, intencion#176, categoria#29
1, urgencia#308, frustracion#326]
Arguments: [tweet_id#0, rol#1, fecha#3, texto_limpio#5, intencion#176, categoria#29
1, urgencia#308, frustracion#326]

(2) InMemoryRelation
Arguments: [tweet_id#0, rol#1, fecha#3, texto_limpio#5, intencion#176, categoria#29
1, urgencia#308, frustracion#326], CachedRDDBuilder(org.apache.spark.sql.execution.c
olumnar.DefaultCachedBatchSerializer@1da35c1,StorageLevel(disk, memory, 1 replicas),
*(1) Project [tweet_id#0, rol#1, fecha#3, texto_limpio#5, intencion#176, CASE WHEN
((_max_score#275 > 0) AND (_score_telefonia_llamadas#183 = _max_score#275)) THEN tel
efonia_llamadas ELSE CASE WHEN ((_max_score#275 > 0) AND (_score_cuenta_icloud#191 =
_max_score#275)) THEN cuenta_icloud ELSE CASE WHEN ((_max_score#275 > 0) AND (_score
_pagos_facturacion#200 = _max_score#275)) THEN pagos_facturacion ELSE CASE WHEN ((_m
ax_score#275 > 0) AND (_score_servicios_apple#210 = _max_score#275)) THEN servicios_
apple ELSE CASE WHEN ((_max_score#275 > 0) AND (_score_conectividad#221 = _max_score
#275)) THEN conectividad ELSE CASE WHEN ((_max_score#275 > 0) AND (_score_apps_funci
ones#233 = _max_score#275)) THEN apps_funciones ELSE CASE WHEN ((_max_score#275 > 0)
AND (_score_ios_software_rendimiento#246 = _max_score#275)) THEN ios_software_rendim
iento ELSE CASE WHEN ((_max_score#275 > 0) AND (_score_hardware_dispositivo#260 = _m
ax_score#275)) THEN hardware_dispositivo ELSE CASE WHEN RLIKE(_t#170, (?i)\b(?:iphon
e|ipad|ipod|mac(?:book)?\b|imac|phone|(?:apple )?watch\b|airpods|device|laptop|table
t)) THEN dispositivo_general ELSE otros END END END END END END END END END AS categ
oria#291, CASE WHEN NOT (intencion#176 = falla) THEN baja WHEN (RLIKE(_t#170, (?i)\b
(?:outage|down for|(?:is|are) down\b|services? (?:is |are )?down|completely (?:dead|
broken|unusable|died)|unusable|bricked|won'?t (?:turn on|power|boot|start)|boot ?loo
p|stuck on (?:the )?(?:apple )?logo|data loss|lost (?:all|everything|my (?:photos|da
ta|contacts|messages))|(?:photos|data|contacts|messages) (?:gone|deleted|disappeared
|lost)|hacked|unauthori[sz]ed|stolen|fraud|locked out|(?:can'?t|cannot|unable to|not
able to)\s+(?:log ?in|sign ?in|access)|charged (?:twice|\d+)|double charged|disabled
|no service|911|emergency|(?:not able to|unable to|can'?t|cannot|couldn'?t)\s+(?:\w+
\s+){0,3}?(?:make|receive)\s+(?:\w+\s+){0,2}?calls?|(?:fell|fallen|dropped|water|we
t)[^.!?]{0,40}(?:stopped working|dead|not working|won'?t turn on))) OR (RLIKE(_t#17
0, (?i)\b(?:not working|stopped working|not able|unable|won'?t|can'?t|cannot|could
n'?t|no longer|broken|crash|bug|error|stuck|freez|frozen|glitch|lag|fail|outage|down
\b|slow|drain|dying|died|dies\b|lost|missing|disappear|black screen|unresponsive|sof
t reset|hard reset|force restart|fallen|fell\b|cracked|overheat|ruin|(?:isn|aren|was
n|weren|doesn|don|didn|hasn|haven|hadn)'?t\s+(?:\w+\s+){0,2}?(?:allow\w*|let\w*|work
\w*|respond\w*|load\w*|show\w*|charg\w*|connect\w*|sync\w*|open\w*|turn\w*)|not\s+
(?:receiving|sending|syncing|charging|loading|showing|allowing|letting|working|respo
nding)|problem|issue|fix|keeps|restart|reboot|reset)) AND RLIKE(_t#170, (?i)\b(?:urg
ent|asap|immediately|still\b|for (?:days|hours|weeks|months)|since (?:yesterday|mond
ay|tuesday|wednesday|thursday|friday|this morning|last)|(?:past|last) (?:\w+ ){0,2}
(?:hours?|days?|weeks?)|twice|(?:three|four|multiple) times|again|constantly|every
(?:time|day|hour)|all the time|everyone|anyone else|all (?:users|of us)|for work|wor
k(?:ing)? from|business)))) THEN critica WHEN RLIKE(_t#170, (?i)\b(?:not working|sto
pped working|not able|unable|won'?t|can'?t|cannot|couldn'?t|no longer|broken|crash|b
ug|error|stuck|freez|frozen|glitch|lag|fail|outage|down\b|slow|drain|dying|died|dies
\b|lost|missing|disappear|black screen|unresponsive|soft reset|hard reset|force rest
art|fallen|fell\b|cracked|overheat|ruin|(?:isn|aren|wasn|weren|doesn|don|didn|hasn|h
aven|hadn)'?t\s+(?:\w+\s+){0,2}?(?:allow\w*|let\w*|work\w*|respond\w*|load\w*|show\w
*|charg\w*|connect\w*|sync\w*|open\w*|turn\w*)|not\s+(?:receiving|sending|syncing|ch
arging|loading|showing|allowing|letting|working|responding)|problem|issue|fix|keeps|
restart|reboot|reset)) THEN media ELSE baja END AS urgencia#308, RLIKE(_t#170, (?i)
\b(?:fuck|shit|damn|crap|ridiculous|unacceptable|worst|horrible|frustrat|angry|switc
h(?:ing)? to|never (?:buy|again)|😡|👿|😤)) AS frustracion#326]
+- *(1) Project [tweet_id#0, rol#1, fecha#3, texto_limpio#5, _t#170, intencion#176,
_score_telefonia_llamadas#183, _score_cuenta_icloud#191, _score_pagos_facturacion#20
```

```
0, _score_servicios_apple#210, _score_conectividad#221, _score_apps_funciones#233, _
score_ios_software_rendimiento#246, _score_hardware_dispositivo#260, greatest(_score
_telefonia_llamadas#183, _score_cuenta_icloud#191, _score_pagos_facturacion#200, _sc
ore_servicios_apple#210, _score_conectividad#221, _score_apps_funciones#233, _score_
ios_software_rendimiento#246, _score_hardware_dispositivo#260) AS _max_score#275]
   +- *(1) Project [tweet_id#0, rol#1, fecha#3, texto_limpio#5, _t#170, CASE WHEN RL
IKE(_t#170, (?i)\b(?:not working|stopped working|not able|unable|won'?t|can'?t|canno
t|couldn'?t|no longer|broken|crash|bug|error|stuck|freez|frozen|glitch|lag|fail|outa
ge|down\b|slow|drain|dying|died|dies\b|lost|missing|disappear|black screen|unrespons
ive|soft reset|hard reset|force restart|fallen|fell\b|cracked|overheat|ruin|(?:isn|a
ren|wasn|weren|doesn|don|didn|hasn|haven|hadn)'?t\s+(?:\w+\s+){0,2}?(?:allow\w*|let
\w*|work\w*|respond\w*|load\w*|show\w*|charg\w*|connect\w*|sync\w*|open\w*|turn\w*)|
not\s+(?:receiving|sending|syncing|charging|loading|showing|allowing|letting|working
|responding)|outage|down for|(?:is|are) down\b|services? (?:is |are )?down|completel
y (?:dead|broken|unusable|died)|unusable|bricked|won'?t (?:turn on|power|boot|start)
|boot ?loop|stuck on (?:the )?(?:apple )?logo|data loss|lost (?:all|everything|my
(?:photos|data|contacts|messages))|(?:photos|data|contacts|messages) (?:gone|deleted
|disappeared|lost)|hacked|unauthori[sz]ed|stolen|fraud|locked out|(?:can'?t|cannot|u
nable to|not able to)\s+(?:log ?in|sign ?in|access)|charged (?:twice|\d+)|double cha
rged|disabled|no service|911|emergency|(?:not able to|unable to|can'?t|cannot|could
n'?t)\s+(?:\w+\s+){0,3}?(?:make|receive)\s+(?:\w+\s+){0,2}?calls?|(?:fell|fallen|dro
pped|water|wet)[^.!?]{0,40}(?:stopped working|dead|not working|won'?t turn on))) THE
N falla WHEN RLIKE(_t#170, (?i)\b(?:how (?:do|can|to|does|should)|is it possible|is
there a way)) THEN solicitud WHEN RLIKE(_t#170, (?i)\b(?:issue|problem|fix|restart|r
eboot|reset|keeps|annoying|weird|wrong)) THEN falla WHEN (RLIKE(_t#170, (?i)\b(?:whe
re (?:do|can|is|are)|what (?:is|does|are|happens)|when (?:will|is|does)|is there|doe
s (?:anyone|apple|it|the)|do (?:you|apple)|can (?:i|you)|could (?:you|i)|any (?:way|
idea|tips?|update|news)|please (?:help|advise)|how much|which|wondering|want to know
|looking for|need (?:help|to)|i want to|would like to|trying to)) OR Contains(_t#17
0, ?)) THEN solicitud ELSE otro END AS intencion#176, ((((((CASE WHEN RLIKE(_t#170,
(?i)\b(?:(?:phone |voice )?calls?\b)) THEN 1 ELSE 0 END + CASE WHEN RLIKE(_t#170, (?
i)\b(?:phone calls?)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:voicemai
l)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:dial)) THEN 1 ELSE 0 END)
+ CASE WHEN RLIKE(_t#170, (?i)\b(?:ringing)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t
#170, (?i)\b(?:no service)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:\b
caller)) THEN 1 ELSE 0 END) AS _score_telefonia_llamadas#183, ((((((((((CASE WHEN RL
IKE(_t#170, (?i)\b(?:icloud)) THEN 1 ELSE 0 END + CASE WHEN RLIKE(_t#170, (?i)\b(?:a
pple id)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:password)) THEN 1 EL
SE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:log ?in)) THEN 1 ELSE 0 END) + CASE WHE
N RLIKE(_t#170, (?i)\b(?:sign ?in)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?
i)\b(?:locked)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:verification))
THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:2fa)) THEN 1 ELSE 0 END) + CAS
E WHEN RLIKE(_t#170, (?i)\b(?:backup)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170,
(?i)\b(?:storage)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:account)) T
HEN 1 ELSE 0 END) AS _score_cuenta_icloud#191, ((((((((((CASE WHEN RLIKE(_t#170, (?
i)\b(?:charged)) THEN 1 ELSE 0 END + CASE WHEN RLIKE(_t#170, (?i)\b(?:charges?\b)) T
HEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:billing)) THEN 1 ELSE 0 END) +
CASE WHEN RLIKE(_t#170, (?i)\b(?:\bbills?\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_
t#170, (?i)\b(?:refund)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:payme
nt)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:subscription)) THEN 1 ELS
E 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:invoice)) THEN 1 ELSE 0 END) + CASE WHEN
RLIKE(_t#170, (?i)\b(?:apple pay)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)
\b(?:credit card)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:purchase))
THEN 1 ELSE 0 END) AS _score_pagos_facturacion#200, ((((((((((CASE WHEN RLIKE(_t#17
0, (?i)\b(?:apple music)) THEN 1 ELSE 0 END + CASE WHEN RLIKE(_t#170, (?i)\b(?:imess
age)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:facetime)) THEN 1 ELSE 0
END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:siri)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE
(_t#170, (?i)\b(?:icloud mail)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b
(?:apple tv)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:podcast)) THEN 1
ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:maps\b)) THEN 1 ELSE 0 END) + CASE WH
EN RLIKE(_t#170, (?i)\b(?:itunes)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)
\b(?:airdrop)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:homepod)) THEN
1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:carplay)) THEN 1 ELSE 0 END) AS _sc
ore_servicios_apple#210, ((((((((((CASE WHEN RLIKE(_t#170, (?i)\b(?:wi-?fi)) THEN
1 ELSE 0 END + CASE WHEN RLIKE(_t#170, (?i)\b(?:bluetooth)) THEN 1 ELSE 0 END) + CAS
E WHEN RLIKE(_t#170, (?i)\b(?:cellular)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#17
0, (?i)\b(?:signal)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:network))
THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:mobile data)) THEN 1 ELSE 0 EN
D) + CASE WHEN RLIKE(_t#170, (?i)\b(?:data\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE
(_t#170, (?i)\b(?:sim\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:carr
ier)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:roaming)) THEN 1 ELSE 0
END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:hotspot)) THEN 1 ELSE 0 END) + CASE WHEN RLI
KE(_t#170, (?i)\b(?:lte\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:in
```

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ternet)) THEN 1 ELSE 0 END) AS _score_conectividad#221, ((((((((((CASE WHEN RLIKE(_
t#170, (?i)\b(?:app store)) THEN 1 ELSE 0 END + CASE WHEN RLIKE(_t#170, (?i)\b(?:app
s?\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:download)) THEN 1 ELSE
0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:testflight)) THEN 1 ELSE 0 END) + CASE WHE
N RLIKE(_t#170, (?i)\b(?:safari)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)
\b(?:mail\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:photos?\b)) THEN
1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:widget)) THEN 1 ELSE 0 END) + CASE
WHEN RLIKE(_t#170, (?i)\b(?:notification)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#1
70, (?i)\b(?:feature)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:quick e
dit)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:edit\b)) THEN 1 ELSE 0 E
ND) AS _score_apps_funciones#233, ((((((((((((((CASE WHEN RLIKE(_t#170, (?i)\b(?:i
os(?:\s?\d+)?\b)) THEN 1 ELSE 0 END + CASE WHEN RLIKE(_t#170, (?i)\b(?:update)) THEN
1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:upgrade)) THEN 1 ELSE 0 END) + CASE
WHEN RLIKE(_t#170, (?i)\b(?:macos)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?
i)\b(?:beta)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:install)) THEN 1
ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:firmware)) THEN 1 ELSE 0 END) + CASE
WHEN RLIKE(_t#170, (?i)\b(?:keyboard)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170,
(?i)\b(?:glitch)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:bug)) THEN 1
ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:freez)) THEN 1 ELSE 0 END) + CASE WHE
N RLIKE(_t#170, (?i)\b(?:frozen)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)
\b(?:crash)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:lag)) THEN 1 ELSE
0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:slow)) THEN 1 ELSE 0 END) + CASE WHEN RLIK
E(_t#170, (?i)\b(?:unresponsive)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)
\b(?:performance)) THEN 1 ELSE 0 END) AS _score_ios_software_rendimiento#246,
((((((((((((((((CASE WHEN RLIKE(_t#170, (?i)\b(?:battery)) THEN 1 ELSE 0 END + CASE
WHEN RLIKE(_t#170, (?i)\b(?:screens?\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#17
0, (?i)\b(?:charging)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:charge
r)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:cracked)) THEN 1 ELSE 0 EN
D) + CASE WHEN RLIKE(_t#170, (?i)\b(?:fallen)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE
(_t#170, (?i)\b(?:fell\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:dro
pped)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:water)) THEN 1 ELSE 0 E
ND) + CASE WHEN RLIKE(_t#170, (?i)\b(?:speaker)) THEN 1 ELSE 0 END) + CASE WHEN RLIK
E(_t#170, (?i)\b(?:microphone)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b
(?:buttons?\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:camera)) THEN
1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:overheat)) THEN 1 ELSE 0 END) + CAS
E WHEN RLIKE(_t#170, (?i)\b(?:turn(?:ing)? on)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE
(_t#170, (?i)\b(?:display)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:to
uch id)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:face id)) THEN 1 ELSE
0 END) AS _score_hardware_dispositivo#260]
      +- *(1) Project [tweet_id#0, rol#1, fecha#3, texto_limpio#5, regexp_replace(co
alesce(texto_limpio#5, ), [’‘`], ', 1) AS _t#170]
         +- *(1) Filter (isnotnull(rol#1) AND (rol#1 = cliente))
            +- *(1) ColumnarToRow
               +- FileScan parquet [tweet_id#0,rol#1,fecha#3,texto_limpio#5] Batche
d: true, DataFilters: [isnotnull(rol#1), (rol#1 = cliente)], Format: Parquet, Locati
on: InMemoryFileIndex(1 paths)[file:/c:/Users/julia/OneDrive/Escritorio/MCD/Tesis/pr
oyecto_investigac..., PartitionFilters: [], PushedFilters: [IsNotNull(rol), EqualTo
(rol,cliente)], ReadSchema: struct<tweet_id:string,rol:string,fecha:timestamp_ntz,te
xto_limpio:string>
,None)

(3) Scan parquet
Output [4]: [tweet_id#0, rol#1, fecha#3, texto_limpio#5]
Batched: true
Location: InMemoryFileIndex [file:/c:/Users/julia/OneDrive/Escritorio/MCD/Tesis/proy
ecto_investigacion/data/processed/apple_support_filtrado.parquet]
PushedFilters: [IsNotNull(rol), EqualTo(rol,cliente)]
ReadSchema: struct<tweet_id:string,rol:string,fecha:timestamp_ntz,texto_limpio:strin
g>

(4) ColumnarToRow [codegen id : 1]
Input [4]: [tweet_id#0, rol#1, fecha#3, texto_limpio#5]

(5) Filter [codegen id : 1]
Input [4]: [tweet_id#0, rol#1, fecha#3, texto_limpio#5]
Condition : (isnotnull(rol#1) AND (rol#1 = cliente))

(6) Project [codegen id : 1]
Output [5]: [tweet_id#0, rol#1, fecha#3, texto_limpio#5, regexp_replace(coalesce(tex
to_limpio#5, ), [’‘`], ', 1) AS _t#170]
Input [4]: [tweet_id#0, rol#1, fecha#3, texto_limpio#5]

(7) Project [codegen id : 1]
```

```
Output [14]: [tweet_id#0, rol#1, fecha#3, texto_limpio#5, _t#170, CASE WHEN RLIKE(_t
#170, (?i)\b(?:not working|stopped working|not able|unable|won'?t|can'?t|cannot|coul
dn'?t|no longer|broken|crash|bug|error|stuck|freez|frozen|glitch|lag|fail|outage|dow
n\b|slow|drain|dying|died|dies\b|lost|missing|disappear|black screen|unresponsive|so
ft reset|hard reset|force restart|fallen|fell\b|cracked|overheat|ruin|(?:isn|aren|wa
sn|weren|doesn|don|didn|hasn|haven|hadn)'?t\s+(?:\w+\s+){0,2}?(?:allow\w*|let\w*|wor
k\w*|respond\w*|load\w*|show\w*|charg\w*|connect\w*|sync\w*|open\w*|turn\w*)|not\s+
(?:receiving|sending|syncing|charging|loading|showing|allowing|letting|working|respo
nding)|outage|down for|(?:is|are) down\b|services? (?:is |are )?down|completely (?:d
ead|broken|unusable|died)|unusable|bricked|won'?t (?:turn on|power|boot|start)|boot
?loop|stuck on (?:the )?(?:apple )?logo|data loss|lost (?:all|everything|my (?:photo
s|data|contacts|messages))|(?:photos|data|contacts|messages) (?:gone|deleted|disappe
ared|lost)|hacked|unauthori[sz]ed|stolen|fraud|locked out|(?:can'?t|cannot|unable to
|not able to)\s+(?:log ?in|sign ?in|access)|charged (?:twice|\d+)|double charged|dis
abled|no service|911|emergency|(?:not able to|unable to|can'?t|cannot|couldn'?t)\s+
(?:\w+\s+){0,3}?(?:make|receive)\s+(?:\w+\s+){0,2}?calls?|(?:fell|fallen|dropped|wat
er|wet)[^.!?]{0,40}(?:stopped working|dead|not working|won'?t turn on))) THEN falla
WHEN RLIKE(_t#170, (?i)\b(?:how (?:do|can|to|does|should)|is it possible|is there a
way)) THEN solicitud WHEN RLIKE(_t#170, (?i)\b(?:issue|problem|fix|restart|reboot|re
set|keeps|annoying|weird|wrong)) THEN falla WHEN (RLIKE(_t#170, (?i)\b(?:where (?:do
|can|is|are)|what (?:is|does|are|happens)|when (?:will|is|does)|is there|does (?:any
one|apple|it|the)|do (?:you|apple)|can (?:i|you)|could (?:you|i)|any (?:way|idea|tip
s?|update|news)|please (?:help|advise)|how much|which|wondering|want to know|looking
for|need (?:help|to)|i want to|would like to|trying to)) OR Contains(_t#170, ?)) THE
N solicitud ELSE otro END AS intencion#176, ((((((CASE WHEN RLIKE(_t#170, (?i)\b(?:
(?:phone |voice )?calls?\b)) THEN 1 ELSE 0 END + CASE WHEN RLIKE(_t#170, (?i)\b(?:ph
one calls?)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:voicemail)) THEN
1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:dial)) THEN 1 ELSE 0 END) + CASE WH
EN RLIKE(_t#170, (?i)\b(?:ringing)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?
i)\b(?:no service)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:\bcaller))
THEN 1 ELSE 0 END) AS _score_telefonia_llamadas#183, ((((((((((CASE WHEN RLIKE(_t#17
0, (?i)\b(?:icloud)) THEN 1 ELSE 0 END + CASE WHEN RLIKE(_t#170, (?i)\b(?:apple id))
THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:password)) THEN 1 ELSE 0 END)
+ CASE WHEN RLIKE(_t#170, (?i)\b(?:log ?in)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t
#170, (?i)\b(?:sign ?in)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:lock
ed)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:verification)) THEN 1 ELS
E 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:2fa)) THEN 1 ELSE 0 END) + CASE WHEN RLI
KE(_t#170, (?i)\b(?:backup)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:s
torage)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:account)) THEN 1 ELSE
0 END) AS _score_cuenta_icloud#191, ((((((((((CASE WHEN RLIKE(_t#170, (?i)\b(?:charg
ed)) THEN 1 ELSE 0 END + CASE WHEN RLIKE(_t#170, (?i)\b(?:charges?\b)) THEN 1 ELSE 0
END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:billing)) THEN 1 ELSE 0 END) + CASE WHEN RLI
KE(_t#170, (?i)\b(?:\bbills?\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b
(?:refund)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:payment)) THEN 1 E
LSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:subscription)) THEN 1 ELSE 0 END) + CA
SE WHEN RLIKE(_t#170, (?i)\b(?:invoice)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#17
0, (?i)\b(?:apple pay)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:credit
card)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:purchase)) THEN 1 ELSE
0 END) AS _score_pagos_facturacion#200, ((((((((((CASE WHEN RLIKE(_t#170, (?i)\b(?:
apple music)) THEN 1 ELSE 0 END + CASE WHEN RLIKE(_t#170, (?i)\b(?:imessage)) THEN 1
ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:facetime)) THEN 1 ELSE 0 END) + CASE
WHEN RLIKE(_t#170, (?i)\b(?:siri)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)
\b(?:icloud mail)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:apple tv))
THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:podcast)) THEN 1 ELSE 0 END) +
CASE WHEN RLIKE(_t#170, (?i)\b(?:maps\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#17
0, (?i)\b(?:itunes)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:airdrop))
THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:homepod)) THEN 1 ELSE 0 END) +
CASE WHEN RLIKE(_t#170, (?i)\b(?:carplay)) THEN 1 ELSE 0 END) AS _score_servicios_ap
ple#210, ((((((((((((CASE WHEN RLIKE(_t#170, (?i)\b(?:wi-?fi)) THEN 1 ELSE 0 END + C
ASE WHEN RLIKE(_t#170, (?i)\b(?:bluetooth)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#
170, (?i)\b(?:cellular)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:signa
l)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:network)) THEN 1 ELSE 0 EN
D) + CASE WHEN RLIKE(_t#170, (?i)\b(?:mobile data)) THEN 1 ELSE 0 END) + CASE WHEN R
LIKE(_t#170, (?i)\b(?:data\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b
(?:sim\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:carrier)) THEN 1 EL
SE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:roaming)) THEN 1 ELSE 0 END) + CASE WHE
N RLIKE(_t#170, (?i)\b(?:hotspot)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)
\b(?:lte\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:internet)) THEN 1
ELSE 0 END) AS _score_conectividad#221, ((((((((((CASE WHEN RLIKE(_t#170, (?i)\b(?:
app store)) THEN 1 ELSE 0 END + CASE WHEN RLIKE(_t#170, (?i)\b(?:apps?\b)) THEN 1 EL
SE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:download)) THEN 1 ELSE 0 END) + CASE WH
EN RLIKE(_t#170, (?i)\b(?:testflight)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170,
(?i)\b(?:safari)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:mail\b)) THE
```

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N 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:photos?\b)) THEN 1 ELSE 0 END) +
CASE WHEN RLIKE(_t#170, (?i)\b(?:widget)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#17
0, (?i)\b(?:notification)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:fea
ture)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:quick edit)) THEN 1 ELS
E 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:edit\b)) THEN 1 ELSE 0 END) AS _score_ap
ps_funciones#233, ((((((((((((((CASE WHEN RLIKE(_t#170, (?i)\b(?:ios(?:\s?\d+)?
\b)) THEN 1 ELSE 0 END + CASE WHEN RLIKE(_t#170, (?i)\b(?:update)) THEN 1 ELSE 0 EN
D) + CASE WHEN RLIKE(_t#170, (?i)\b(?:upgrade)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE
(_t#170, (?i)\b(?:macos)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:bet
a)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:install)) THEN 1 ELSE 0 EN
D) + CASE WHEN RLIKE(_t#170, (?i)\b(?:firmware)) THEN 1 ELSE 0 END) + CASE WHEN RLIK
E(_t#170, (?i)\b(?:keyboard)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:
glitch)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:bug)) THEN 1 ELSE 0 E
ND) + CASE WHEN RLIKE(_t#170, (?i)\b(?:freez)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE
(_t#170, (?i)\b(?:frozen)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:cra
sh)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:lag)) THEN 1 ELSE 0 END)
+ CASE WHEN RLIKE(_t#170, (?i)\b(?:slow)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#17
0, (?i)\b(?:unresponsive)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:per
formance)) THEN 1 ELSE 0 END) AS _score_ios_software_rendimiento#246,
((((((((((((((((CASE WHEN RLIKE(_t#170, (?i)\b(?:battery)) THEN 1 ELSE 0 END + CASE
WHEN RLIKE(_t#170, (?i)\b(?:screens?\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#17
0, (?i)\b(?:charging)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:charge
r)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:cracked)) THEN 1 ELSE 0 EN
D) + CASE WHEN RLIKE(_t#170, (?i)\b(?:fallen)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE
(_t#170, (?i)\b(?:fell\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:dro
pped)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:water)) THEN 1 ELSE 0 E
ND) + CASE WHEN RLIKE(_t#170, (?i)\b(?:speaker)) THEN 1 ELSE 0 END) + CASE WHEN RLIK
E(_t#170, (?i)\b(?:microphone)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b
(?:buttons?\b)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:camera)) THEN
1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:overheat)) THEN 1 ELSE 0 END) + CAS
E WHEN RLIKE(_t#170, (?i)\b(?:turn(?:ing)? on)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE
(_t#170, (?i)\b(?:display)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:to
uch id)) THEN 1 ELSE 0 END) + CASE WHEN RLIKE(_t#170, (?i)\b(?:face id)) THEN 1 ELSE
0 END) AS _score_hardware_dispositivo#260]
Input [5]: [tweet_id#0, rol#1, fecha#3, texto_limpio#5, _t#170]

(8) Project [codegen id : 1]
Output [15]: [tweet_id#0, rol#1, fecha#3, texto_limpio#5, _t#170, intencion#176, _sc
ore_telefonia_llamadas#183, _score_cuenta_icloud#191, _score_pagos_facturacion#200,
_score_servicios_apple#210, _score_conectividad#221, _score_apps_funciones#233, _sco
re_ios_software_rendimiento#246, _score_hardware_dispositivo#260, greatest(_score_te
lefonia_llamadas#183, _score_cuenta_icloud#191, _score_pagos_facturacion#200, _score
_servicios_apple#210, _score_conectividad#221, _score_apps_funciones#233, _score_ios
_software_rendimiento#246, _score_hardware_dispositivo#260) AS _max_score#275]
Input [14]: [tweet_id#0, rol#1, fecha#3, texto_limpio#5, _t#170, intencion#176, _sco
re_telefonia_llamadas#183, _score_cuenta_icloud#191, _score_pagos_facturacion#200, _
score_servicios_apple#210, _score_conectividad#221, _score_apps_funciones#233, _scor
e_ios_software_rendimiento#246, _score_hardware_dispositivo#260]

(9) Project [codegen id : 1]
Output [8]: [tweet_id#0, rol#1, fecha#3, texto_limpio#5, intencion#176, CASE WHEN
((_max_score#275 > 0) AND (_score_telefonia_llamadas#183 = _max_score#275)) THEN tel
efonia_llamadas ELSE CASE WHEN ((_max_score#275 > 0) AND (_score_cuenta_icloud#191 =
_max_score#275)) THEN cuenta_icloud ELSE CASE WHEN ((_max_score#275 > 0) AND (_score
_pagos_facturacion#200 = _max_score#275)) THEN pagos_facturacion ELSE CASE WHEN ((_m
ax_score#275 > 0) AND (_score_servicios_apple#210 = _max_score#275)) THEN servicios_
apple ELSE CASE WHEN ((_max_score#275 > 0) AND (_score_conectividad#221 = _max_score
#275)) THEN conectividad ELSE CASE WHEN ((_max_score#275 > 0) AND (_score_apps_funci
ones#233 = _max_score#275)) THEN apps_funciones ELSE CASE WHEN ((_max_score#275 > 0)
AND (_score_ios_software_rendimiento#246 = _max_score#275)) THEN ios_software_rendim
iento ELSE CASE WHEN ((_max_score#275 > 0) AND (_score_hardware_dispositivo#260 = _m
ax_score#275)) THEN hardware_dispositivo ELSE CASE WHEN RLIKE(_t#170, (?i)\b(?:iphon
e|ipad|ipod|mac(?:book)?\b|imac|phone|(?:apple )?watch\b|airpods|device|laptop|table
t)) THEN dispositivo_general ELSE otros END END END END END END END END END AS categ
oria#291, CASE WHEN NOT (intencion#176 = falla) THEN baja WHEN (RLIKE(_t#170, (?i)\b
(?:outage|down for|(?:is|are) down\b|services? (?:is |are )?down|completely (?:dead|
broken|unusable|died)|unusable|bricked|won'?t (?:turn on|power|boot|start)|boot ?loo
p|stuck on (?:the )?(?:apple )?logo|data loss|lost (?:all|everything|my (?:photos|da
ta|contacts|messages))|(?:photos|data|contacts|messages) (?:gone|deleted|disappeared
|lost)|hacked|unauthori[sz]ed|stolen|fraud|locked out|(?:can'?t|cannot|unable to|not
able to)\s+(?:log ?in|sign ?in|access)|charged (?:twice|\d+)|double charged|disabled
|no service|911|emergency|(?:not able to|unable to|can'?t|cannot|couldn'?t)\s+(?:\w+
\s+){0,3}?(?:make|receive)\s+(?:\w+\s+){0,2}?calls?|(?:fell|fallen|dropped|water|we
```

```
t)[^.!?]{0,40}(?:stopped working|dead|not working|won'?t turn on))) OR (RLIKE(_t#17
0, (?i)\b(?:not working|stopped working|not able|unable|won'?t|can'?t|cannot|could
n'?t|no longer|broken|crash|bug|error|stuck|freez|frozen|glitch|lag|fail|outage|down
\b|slow|drain|dying|died|dies\b|lost|missing|disappear|black screen|unresponsive|sof
t reset|hard reset|force restart|fallen|fell\b|cracked|overheat|ruin|(?:isn|aren|was
n|weren|doesn|don|didn|hasn|haven|hadn)'?t\s+(?:\w+\s+){0,2}?(?:allow\w*|let\w*|work
\w*|respond\w*|load\w*|show\w*|charg\w*|connect\w*|sync\w*|open\w*|turn\w*)|not\s+
(?:receiving|sending|syncing|charging|loading|showing|allowing|letting|working|respo
nding)|problem|issue|fix|keeps|restart|reboot|reset)) AND RLIKE(_t#170, (?i)\b(?:urg
ent|asap|immediately|still\b|for (?:days|hours|weeks|months)|since (?:yesterday|mond
ay|tuesday|wednesday|thursday|friday|this morning|last)|(?:past|last) (?:\w+ ){0,2}
(?:hours?|days?|weeks?)|twice|(?:three|four|multiple) times|again|constantly|every
(?:time|day|hour)|all the time|everyone|anyone else|all (?:users|of us)|for work|wor
k(?:ing)? from|business)))) THEN critica WHEN RLIKE(_t#170, (?i)\b(?:not working|sto
pped working|not able|unable|won'?t|can'?t|cannot|couldn'?t|no longer|broken|crash|b
ug|error|stuck|freez|frozen|glitch|lag|fail|outage|down\b|slow|drain|dying|died|dies
\b|lost|missing|disappear|black screen|unresponsive|soft reset|hard reset|force rest
art|fallen|fell\b|cracked|overheat|ruin|(?:isn|aren|wasn|weren|doesn|don|didn|hasn|h
aven|hadn)'?t\s+(?:\w+\s+){0,2}?(?:allow\w*|let\w*|work\w*|respond\w*|load\w*|show\w
*|charg\w*|connect\w*|sync\w*|open\w*|turn\w*)|not\s+(?:receiving|sending|syncing|ch
arging|loading|showing|allowing|letting|working|responding)|problem|issue|fix|keeps|
restart|reboot|reset)) THEN media ELSE baja END AS urgencia#308, RLIKE(_t#170, (?i)
\b(?:fuck|shit|damn|crap|ridiculous|unacceptable|worst|horrible|frustrat|angry|switc
h(?:ing)? to|never (?:buy|again)|🤬|😡|😤)) AS frustracion#326]
Input [15]: [tweet_id#0, rol#1, fecha#3, texto_limpio#5, _t#170, intencion#176, _sco
re_telefonia_llamadas#183, _score_cuenta_icloud#191, _score_pagos_facturacion#200, _
score_servicios_apple#210, _score_conectividad#221, _score_apps_funciones#233, _scor
e_ios_software_rendimiento#246, _score_hardware_dispositivo#260, _max_score#275]

(10) AdaptiveSparkPlan
Output [8]: [tweet_id#0, rol#1, fecha#3, texto_limpio#5, intencion#176, categoria#29
1, urgencia#308, frustracion#326]
Arguments: isFinalPlan=false


Total de tweets de clientes etiquetados: 126,113

=== DISTRIBUCIÓN: INTENCION ===
+---------+-----+-----+
|intencion|count|pct  |
+---------+-----+-----+
|falla    |63532|0.504|
|otro     |41434|0.329|
|solicitud|21147|0.168|
+---------+-----+-----+


=== DISTRIBUCIÓN: CATEGORIA ===
+------------------------+-----+-----+
|categoria               |count|pct  |
+------------------------+-----+-----+
|otros                   |38867|0.308|
|ios_software_rendimiento|38407|0.305|
|dispositivo_general     |13224|0.105|
|apps_funciones          |9367 |0.074|
|hardware_dispositivo    |6982 |0.055|
|cuenta_icloud           |4929 |0.039|
|servicios_apple         |4397 |0.035|
|conectividad            |4267 |0.034|
|pagos_facturacion       |3253 |0.026|
|telefonia_llamadas      |2420 |0.019|
+------------------------+-----+-----+


=== DISTRIBUCIÓN: URGENCIA ===
+--------+-----+-----+
|urgencia|count|pct  |
+--------+-----+-----+
|baja    |64379|0.51 |
|media   |50321|0.399|
|critica |11413|0.09 |
+--------+-----+-----+
```

```
=== DISTRIBUCIÓN: FRUSTRACION ===
+-----------+------+-----+
|frustracion|count |pct  |
+-----------+------+-----+
|false      |113723|0.902|
|true       |12390 |0.098|
+-----------+------+-----+


=== TABLA CRUZADA: CATEGORÍA vs URGENCIA (SOLO FALLAS) ===
+------------------------+----+-----+-------+
|categoria               |baja|media|critica|
+------------------------+----+-----+-------+
|apps_funciones          |130 |4011 |984    |
|conectividad            |53  |1744 |434    |
|cuenta_icloud           |64  |1651 |816    |
|dispositivo_general     |296 |4453 |1035   |
|hardware_dispositivo    |86  |2894 |681    |
|ios_software_rendimiento|383 |21077|4711   |
|otros                   |680 |10512|1513   |
|pagos_facturacion       |25  |1117 |307    |
|servicios_apple         |63  |1892 |406    |
|telefonia_llamadas      |18  |970  |526    |
+------------------------+----+-----+-------+
```

# 6. Guardado en Parquet y diagnóstico de cobertura de categorías

Llevamos el resultado etiquetado al Driver con `toPandas()` para almacenarlo en Parquet con PyArrow y revisamos los n-gramas más frecuentes en las fallas sin categoría específica ( `otros` o `dispositivo_general` ).

In [7]:
```
# ============================================================
# 6. GUARDADO EN PARQUET Y REVISIÓN DE COBERTURA
# ============================================================
cl = df_etiquetado.toPandas()
RUTA_OUT.parent.mkdir(parents=True, exist_ok=True)
cl.to_parquet(RUTA_OUT, index=False, engine="pyarrow", compression="snappy")
print("Guardado en:", RUTA_OUT.resolve(), "| Filas:", len(cl))

fallas = cl[cl["intencion"] == "falla"]
sin_tema = fallas["categoria"].isin(["otros", "dispositivo_general"])
print(f"Fallas sin tema específico ('otros' + 'dispositivo_general'): {sin_tema.mea

vec = CountVectorizer(stop_words="english", ngram_range=(1, 2), min_df=20)
X = vec.fit_transform(fallas.loc[sin_tema, "texto_limpio"])
top = sorted(zip(vec.get_feature_names_out(), X.sum(axis=0).A1), key=lambda x: -x[1
display(pd.DataFrame(top, columns=["termino", "frecuencia"]))
```

```
Guardado en: C:\Users\julia\OneDrive\Escritorio\MCD\Tesis\proyecto_investigacion\dat
a\processed\clientes_etiquetados.parquet | Filas: 126113
Fallas sin tema específico ('otros' + 'dispositivo_general'): 29.1%
```

| | termino | frecuencia |
|---|---|---|
| **0** | user | 23177 |
| **1** | fix | 6487 |
| **2** | user user | 3689 |
| **3** | phone | 2760 |
| **4** | url | 2652 |
| **5** | user fix | 2510 |
| **6** | iphone | 2209 |
| **7** | problem | 2101 |
| **8** | issue | 2006 |
| **9** | shit | 1532 |
| **10** | just | 1378 |
| **11** | apple | 1313 |
| **12** | letter | 1126 |
| **13** | help | 1104 |
| **14** | work | 1058 |
| **15** | type | 909 |
| **16** | annoying | 892 |
| **17** | fixed | 883 |
| **18** | like | 837 |
| **19** | working | 829 |
| **20** | question | 808 |
| **21** | going | 798 |
| **22** | need | 797 |
| **23** | time | 754 |
| **24** | new | 744 |
| **25** | fix shit | 729 |
| **26** | don | 712 |
| **27** | having | 704 |
| **28** | hey | 692 |
| **29** | ve | 682 |

# 7. Muestra estratificada para etiquetado manual (a ciegas)

Extraemos hasta 40 tweets por combinación `intencion × urgencia` y exportamos `muestra_para_etiquetar.csv` (sin etiquetas automáticas) y `muestra_clave_reglas.csv` (con las etiquetas de las reglas).

In [8]:
```
# ============================================================
# 7. GENERACIÓN DE MUESTRA PARA ETIQUETADO MANUAL
# ============================================================
muestra = (
    cl.sample(frac=1, random_state=42)
    .groupby(["intencion", "urgencia"], group_keys=False)
    .head(40)
```

```
    .copy()
)

para_etiquetar = muestra[["tweet_id", "texto_limpio"]].copy()
for c in ["intencion_manual", "categoria_manual", "urgencia_manual", "comentario_er
    para_etiquetar[c] = ""

RUTA_MUESTRA.parent.mkdir(parents=True, exist_ok=True)
para_etiquetar.to_csv(RUTA_MUESTRA, index=False, encoding="utf-8-sig")
muestra[["tweet_id", "intencion", "categoria", "urgencia"]].to_csv(RUTA_CLAVE, inde
print(len(muestra), "tweets exportados en la muestra:", RUTA_MUESTRA.resolve())
```

```
200 tweets exportados en la muestra: C:\Users\julia\OneDrive\Escritorio\MCD\Tesis\pr
oyecto_investigacion\data\processed\muestra_para_etiquetar.csv
```

# 8. Evaluación contra etiquetado manual y cierre de Spark

Una vez completado `muestra_para_etiquetar.csv`, esta celda calcula precisión, recall y **F1-score** por dimensión. Finalmente liberamos la caché y detenemos la `SparkSession`.

In [ ]:
```
# ============================================================
# 8. EVALUACIÓN F1-SCORE Y CIERRE DE SPARK
# ============================================================
manual = pd.read_csv(RUTA_MUESTRA, dtype={"tweet_id": str}).fillna("")
clave = pd.read_csv(RUTA_CLAVE, dtype={"tweet_id": str})
m = manual.merge(clave, on="tweet_id")
m = m[m["intencion_manual"].str.strip() != ""]

if m.empty:
    print("Aún no hay etiquetas manuales en muestra_para_etiquetar.csv.")
else:
    for dim in ["intencion", "categoria", "urgencia"]:
        y_true = m[f"{dim}_manual"].str.strip().str.lower()
        y_pred = m[dim].str.strip().str.lower()
        print(f"\n=== {dim.upper()} (n={len(m)}) ===")
        print(classification_report(y_true, y_pred, digits=3, zero_division=0))

df_etiquetado.unpersist(blocking=True)
spark.stop()
print("SparkSession detenida.")
```